

ERC Marsyard Rover Module-Based Benchmarking Criteria

Benchmark structure divided into Perception, SLAM/Mapping, Path Planning, Control, and Mission Evaluation for a ROS 2 Marsyard rover testing system.

Purpose

This document defines how to evaluate the complete Marsyard rover navigation system using clear module-based benchmarks. Each technical module receives its own measurable score, then all scores are combined into a full system benchmark. The structure is designed for a testing team that already has a Marsyard world and rover model and needs objective testing criteria.

Top-Level System Flow

Scenario YAML -> Marsyard simulation -> Rover sensors -> Perception -> SLAM/Mapping -> Path Planning -> Control -> Mission Supervisor -> Evaluator -> Final Report

Core Module Weights

Module	Weight	Reason
Perception	20%	The rover must correctly observe obstacles, terrain, and sensor data before navigation can be trusted.
SLAM / Mapping	20%	The system must localize the rover and maintain maps/heightmaps/costmaps aligned with the world.
Path Planning	25%	The planner decides the route through checkpoints while avoiding unsafe terrain.
Control	25%	The controller executes the plan safely, smoothly, and without instability.
Mission / Evaluation	10%	The benchmark must verify checkpoint completion, logging, scoring, and repeatable execution.

Full System Score = $0.20 \times \text{Perception} + 0.20 \times \text{SLAM/Mapping} + 0.25 \times \text{Path Planning} + 0.25 \times \text{Control} + 0.10 \times \text{Mission/Evaluation}$

1. Perception Benchmark

Perception is responsible for converting raw sensor data into useful information for navigation. In the Marsyard system, perception should detect obstacles, identify risky terrain, validate sensor frames, and provide data that can support heightmap, traversability, and costmap generation.

Inputs

- /scan or /pointcloud
- /camera/depth or stereo point cloud if available
- /imu
- /tf
- Marsyard world ground truth for validation

Outputs

- Detected obstacles
- Terrain point cloud
- Obstacle / terrain classification
- Sensor validity report

- Optional local elevation/height observations

What is calculated

Metric	How to Calculate	Target
Sensor Availability	$\text{available_required_topics} / \text{total_required_topics} \times 100$	$\geq 95\%$
Detection Recall	$\text{true_detected_obstacles} / \text{total_ground_truth_obstacles} \times 100$	$\geq 85\%$
False Positive Rate	$\text{false_obstacles} / \text{total_detections} \times 100$	$\leq 10\%$
Frame Alignment Error	average transform/position error between sensor data and robot/world frame	$\leq 0.15 \text{ m}$
Perception Latency	time between sensor input and processed output	$\leq 200 \text{ ms}$

Perception Score

Perception Score = $0.20 \times \text{SensorAvailability} + 0.30 \times \text{DetectionRecall} + 0.15 \times \text{FalsePositiveScore} + 0.20 \times \text{FrameAlignmentScore} + 0.15 \times \text{LatencyScore}$

Pass condition: Perception Score $\geq 85/100$.

2. SLAM / Mapping Benchmark

SLAM/Mapping is responsible for estimating the rover pose and building or maintaining the map used by navigation. If the system uses a static Marsyard map, this module still measures localization accuracy, map alignment, drift, and heightmap/costmap consistency.

Inputs

- /odom
- /imu
- /tf
- LiDAR/depth data
- static map or heightmap
- ground truth pose from simulation if available

Outputs

- Estimated rover pose
- Map -> odom -> base_link transform
- Updated map or local map
- Heightmap/costmap alignment result
- Localization confidence/status

What is calculated

Metric	How to Calculate	Target
Pose Error	distance between estimated pose and ground truth pose	$\leq 0.30 \text{ m avg}$
Yaw Error	absolute yaw difference between estimated and ground truth pose	$\leq 5 \text{ deg avg}$
Drift Rate	pose error growth per meter travelled or per minute	low/stable
Map Alignment Error	difference between map features and world/heightmap features	$\leq 0.20 \text{ m}$
Localization Dropouts	number of times localization becomes invalid	0 preferred

Metric	How to Calculate	Target
Map Update Rate	updated map / pose rate during run	stable and sufficient

SLAM / Mapping Score

SLAM Score = $0.30 \times \text{PoseAccuracy} + 0.15 \times \text{YawAccuracy} + 0.20 \times \text{DriftScore} + 0.20 \times \text{MapAlignment} + 0.10 \times \text{Continuity} + 0.05 \times \text{UpdateRate}$

Pass condition: SLAM / Mapping Score $\geq 85/100$.

3. Path Planning Benchmark

Path planning is responsible for creating a valid route from the rover position to the current checkpoint or finish point. In this Marsyard benchmark, the planner must consider obstacles, rough terrain, slopes, and the costmap produced from heightmap/traversability data.

Inputs

- Current rover pose
- Checkpoint or finish goal
- Global costmap
- Traversability map
- Robot footprint and limits
- Reference path for evaluation

Outputs

- Global path
- Path cost
- Planning status
- Replanned path when blocked

What is calculated

Metric	How to Calculate	Target
Planning Success Rate	$\text{successful_plans} / \text{requested_plans} \times 100$	$\geq 90\%$
Planning Time	time required to generate a valid path	≤ 2 sec preferred
Blocked Cell Intersections	number of path points inside lethal/blocked costmap cells	0
Average Path Cost	mean cost value along generated path	as low as possible
Path Length Ratio	$\text{planned_path_length} / \text{reference_path_length}$	≤ 1.35
Replanning Success	$\text{successful_replans} / \text{required_replans} \times 100$	$\geq 85\%$

Path Planning Score

Planning Score = $0.25 \times \text{PlanningSuccess} + 0.15 \times \text{PlanningTimeScore} + 0.25 \times \text{ObstacleAvoidance} + 0.15 \times \text{PathCostScore} + 0.10 \times \text{PathLengthScore} + 0.10 \times \text{ReplanningScore}$

Pass condition: Path Planning Score $\geq 85/100$.

4. Control Benchmark

Control is responsible for converting the planned path into motion commands. The controller should track the path, avoid local obstacles, keep the rover stable, and generate smooth /cmd_vel commands.

Inputs

- Global path
- Local costmap
- Current rover pose
- Odometry
- IMU
- Robot velocity/acceleration limits

Outputs

- /cmd_vel
- Path tracking behavior
- Recovery or stop behavior when unsafe

What is calculated

Metric	How to Calculate	Target
Tracking Error	average distance between rover trajectory and planned path	≤ 0.50 m
Velocity Smoothness	penalty from high acceleration/jerk or unstable commands	smooth commands
Oscillation Count	number of repeated left/right or forward/backward oscillations	low/0
Stuck Events	time windows where commanded motion exists but rover barely moves	0 preferred
Collision Count	number of physical collisions	0
Stability Limit Violations	roll/pitch exceeding scenario limits	0

Control Score

Control Score = $0.25 \times \text{TrackingScore} + 0.15 \times \text{SmoothnessScore} + 0.15 \times \text{OscillationScore} + 0.15 \times \text{StuckAvoidance} + 0.20 \times \text{SafetyScore} + 0.10 \times \text{StabilityScore}$

Pass condition: Control Score $\geq 85/100$.

5. Mission / Evaluation Benchmark

Mission/Evaluation verifies that the benchmark runner works correctly. It checks that checkpoints are sent in order, finish is detected, logs are saved, and final scoring is reproducible across trials.

Inputs

- Scenario YAML
- Checkpoint list
- Rover pose and trajectory
- Collision status
- Time
- Module scores

Outputs

- Checkpoint progress
- Final mission status
- CSV/JSON logs

- Scenario score
- Batch report

What is calculated

Metric	How to Calculate	Target
Checkpoint Completion	$\text{reached_checkpoints} / \text{total_checkpoints} \times 100$	$\geq 80\%$
Finish Completion	finish reached: 100, not reached: 0	100 preferred
Timeout Status	$\text{actual_time} \leq \text{timeout_sec}$	true
Logging Completeness	$\text{valid_output_fields} / \text{required_output_fields} \times 100$	$\geq 95\%$
Batch Reliability	$\text{completed_trials} / \text{requested_trials} \times 100$	$\geq 95\%$

Mission / Evaluation Score

Mission Score = $0.35 \times \text{CheckpointCompletion} + 0.20 \times \text{FinishCompletion} + 0.15 \times \text{TimeoutScore} + 0.15 \times \text{LoggingCompleteness} + 0.15 \times \text{BatchReliability}$

Pass condition: Mission / Evaluation Score $\geq 90/100$.

6. Per-Scenario Benchmark Score

Each scenario should generate one final score based on the five module scores. This makes it easy to compare routes such as right-side checkpoints, left-side checkpoints, center route, slope route, rough-terrain route, and full ERC-like route.

Scenario Score = $0.20 \times \text{Perception} + 0.20 \times \text{SLAM/Mapping} + 0.25 \times \text{PathPlanning} + 0.25 \times \text{Control} + 0.10 \times \text{Mission/Evaluation}$

Example

Module	Score
Perception	88
SLAM / Mapping	84
Path Planning	90
Control	82
Mission / Evaluation	95

Scenario Score = $0.20 \times 88 + 0.20 \times 84 + 0.25 \times 90 + 0.25 \times 82 + 0.10 \times 95 = 86.9 / 100$

7. Full System Benchmark

The full system benchmark is calculated over all scenarios and trials. It should summarize the performance of the complete simulation/testing framework, not only a single run.

Scenario Performance Score = $\text{average}(\text{all scenario scores})$

Full System Score = average scenario performance, with failure-rate penalties applied if needed.

Recommended penalties:

Condition	Penalty
Collision Rate > 10%	-5 points
Timeout Rate > 20%	-5 points
Average Checkpoint Completion < 80%	-10 points
Batch Reliability < 90%	-5 points

8. Final Report Format

Field	Description
Total Scenarios	Number of scenario YAML files executed
Total Trials	Number of repeated runs across all scenarios
Average Perception Score	Mean perception score across all runs
Average SLAM Score	Mean SLAM/mapping score across all runs
Average Planning Score	Mean path planning score across all runs
Average Control Score	Mean control score across all runs
Average Mission Score	Mean mission/evaluation score across all runs
Success Rate	successful runs / total runs
Collision Rate	collided runs / total runs
Timeout Rate	timeout runs / total runs
Full System Score	Average scenario score after penalties

9. Benchmark Grades

Score Range	Grade
90 - 100	Excellent
80 - 89	Good
70 - 79	Acceptable
60 - 69	Weak
Below 60	Failed

10. Minimum Acceptance Criteria

- Perception Score ≥ 85
- SLAM / Mapping Score ≥ 85
- Path Planning Score ≥ 85
- Control Score ≥ 85
- Mission / Evaluation Score ≥ 90
- Full System Score ≥ 80
- Collision Rate $\leq 10\%$
- Timeout Rate $\leq 20\%$
- Average Checkpoint Completion $\geq 80\%$

Core Statement

This benchmark evaluates the Marsyard rover navigation system by separating the system into Perception, SLAM/Mapping, Path Planning, Control, and Mission Evaluation. Each module has its own measurable score, and the final benchmark combines all module scores across all scenarios to produce one complete system score.